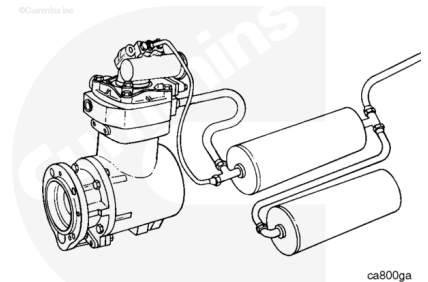


## Trainings ▾

### General Information

The compressed air system normally consists of a gear-driven air compressor, an air governor, air tanks and all necessary plumbing.

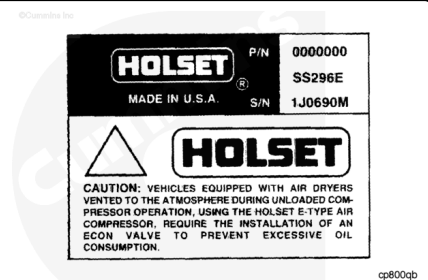
The Holset® single-air compressors are engine-driven, piston-type compressors that supply compressed air to operate air-activated devices. The compressor runs continuously, but has a loaded and unloaded operating mode.



#### E-Type Air Compressor

Holset® SS model air compressors built with the E-Type unloader can be identified by the letter "E" (SS296E, and SS338E) and by the caution on the dataplate.

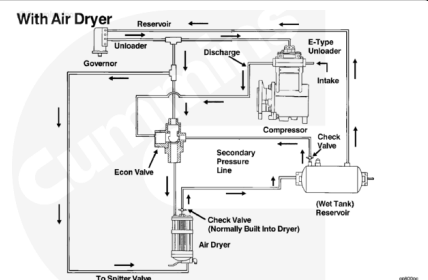
All QE (QE296 and QE338) model air compressors are equipped with the E-Type unloader.



#### E-Type System with Air Dryer

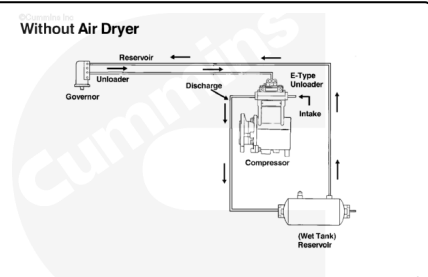
Vehicles equipped with air dryers vented to the atmosphere during unloaded compressor operation, using the Holset® E-Type air compressor, require the installation of an econ valve to prevent excessive oil consumption.

**Note :** Some air dryers can have a built-in econ valve. Check with the manufacturer as to which type is installed.



#### E-Type System without Air Dryer

Air systems **without** air dryers, or with air dryers **not** vented to the atmosphere during unloaded compressor operation, can use the Holset E-Type unloader valve without modifying the air system.

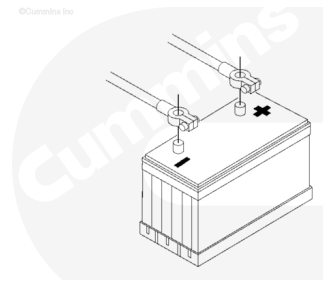


## Preparatory Steps

### **⚠ WARNING ⚠**

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



13900050

### **⚠ WARNING ⚠**

Coolant is toxic. Keep away from pets and children. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

### **⚠ WARNING ⚠**

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

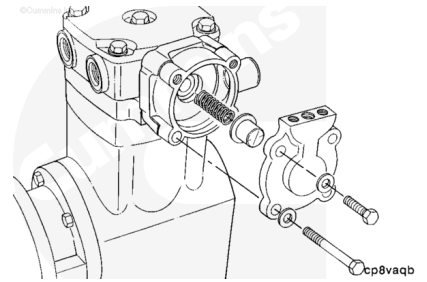
**Note :** If the cylinder head is removed while the air compressor is on the engine, drain the engine coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>).

**Note :** Since the valve plate, head, and unloader body are indexible, marking these parts is recommended to make sure they are reassembled in the proper orientation.

## Disassemble

### **⚠ WARNING ⚠**

The unloader body is installed with spring tension. Use care when removing to prevent personal injury. Always wear protective eyewear.



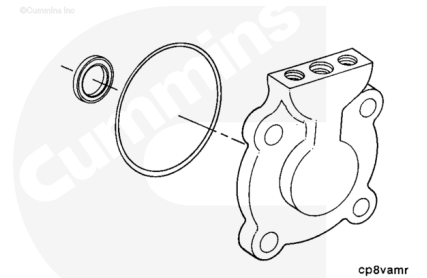
Hold the unloader valve body down, and remove the four cap screws.

Remove the unloader valve body.

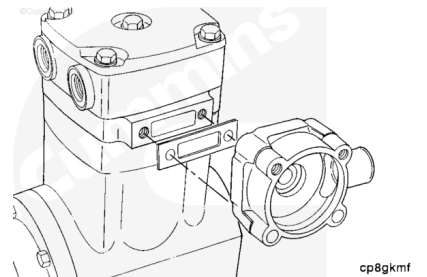
Remove the unloader valve spring.

Remove the unloader valve cap.

Remove the unloader body gasket and unloader valve cap rectangular ring seal.

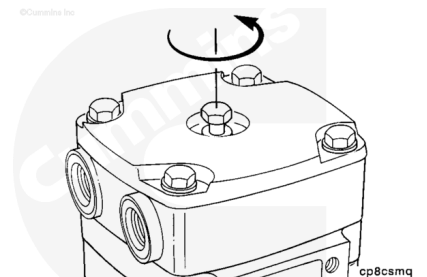


Remove the compressor intake manifold and gasket.

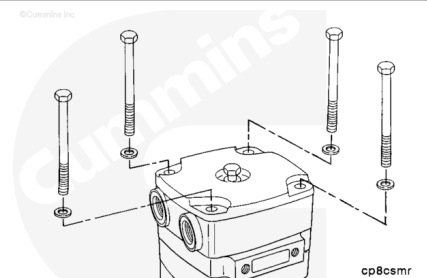


Loosen, but do **not** remove, the center head cap screw.

Mark the head for orientation during assembly.

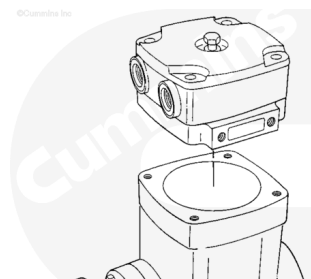


Loosen and remove the four corner head cap screws. Save the cap screws for reuse.



Remove the head, cover, and valve plate assembly and place it on a clean work surface with the intake valve facing upward.

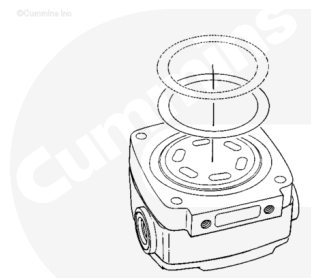
**Note :** If continuing with disassembly of the head, valve plate, and cover, be sure the work surface is clean. Grit pushed into the valve sealing surfaces by setting components on a dirty surface will cause a malfunction after assembly.



cp8hdmg

Some units have a press-fit intake valve retainer. If present, carefully remove it to prevent part damage.

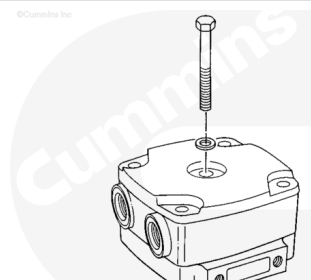
Remove the intake valve.



cp8vams

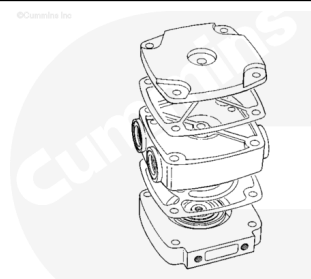
Turn the head assembly over and set it on a clean surface. Remove the center capscrew. This capscrew can be reused.

**Note :** The center capscrew is shorter than the four corner capscrews.



cp8hdec

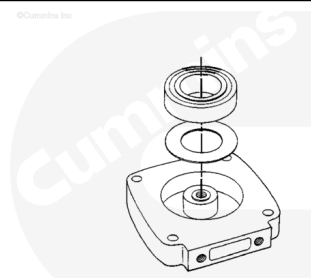
Remove the cover, cover gasket, head, and head gasket.



cp8cvmg

Remove the exhaust valve retainer and exhaust valve.

**Note :** The wave washers must be replaced.

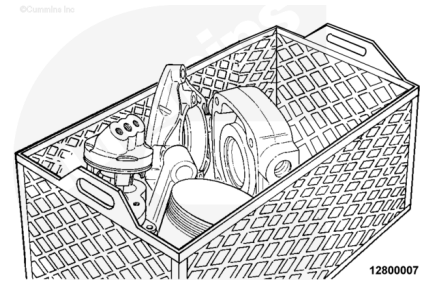


12900076

## Clean and Inspect for Reuse

### **⚠ WARNING ⚠**

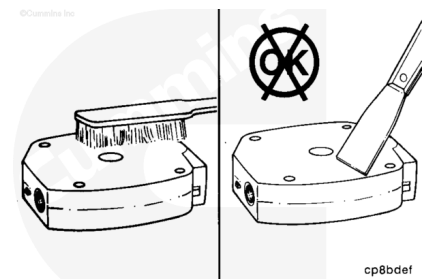
**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing.**



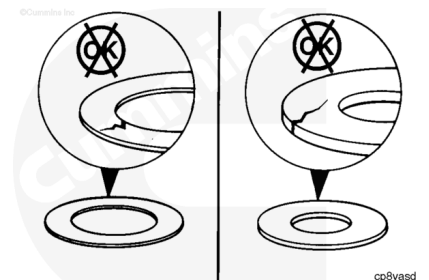
Soak the parts in a kerosene emulsion-based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier can be contacted about solution concentration, temperature, and soak time.

**Note : Do not use a scraper to remove carbon and scale; the sealing surfaces can be damaged.**

The parts can be scrubbed with a stiff, nonmetallic bristle brush.



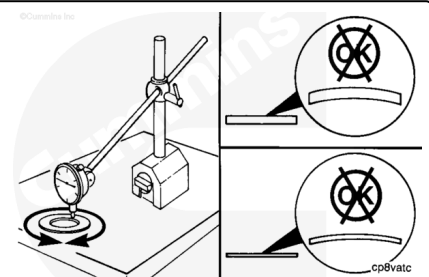
Inspect the intake and exhaust valves for cracks or damage.



Measure the flatness of the intake and exhaust valves. Both valves **must** be flat within 0.03 mm [0.001 in].

Replace valves if cracked, damaged, or **not** flat.

**Note : Holset® Engineering Co., Inc., recommends new valves be installed.**



Inspect the upper part of the unloader valve cap where the rectangular v-seal operates. Check for scoring.

OK



Not OK



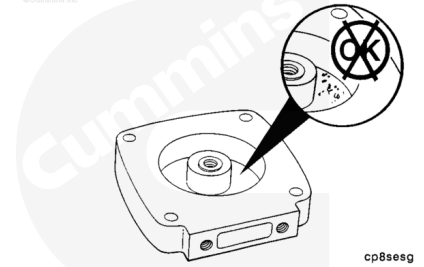
cp8vaasp

Inspect the valve seat surfaces.

**Note :** Inspection of the valve seats in the valve plate requires specialized equipment and is beyond the scope of field service.

If the valve seat is visibly damaged, or can **not** be cleaned, a new valve plate is available in a service kit. Otherwise, a QE valve plate service assembly can be used.

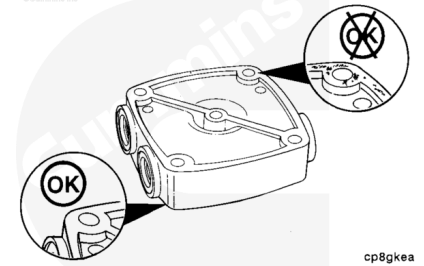
©Cummins Inc.



cp8sesg

Gasket sealing surfaces **must** be clean and free of all old gasket material, carbon, rust, and other buildup. Surfaces **must** be free of scratches, gouges, burrs, and other deformities.

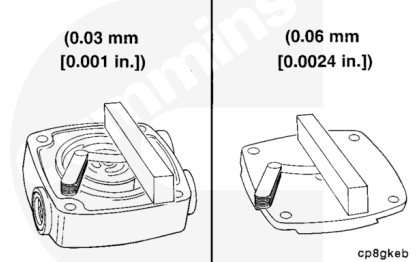
©Cummins Inc.



cp8gkea

After making sure all gasket surfaces are clean and free of the above, inspect the head and cover for flatness. Use the flat plate and the feeler gauges.

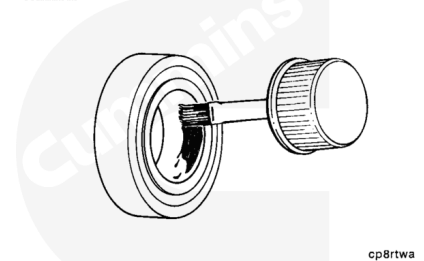
©Cummins Inc.



cp8gkeb

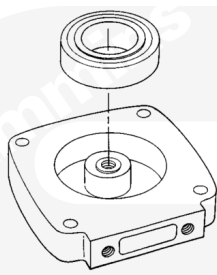
Apply a thin coating of anti-seize compound to the inside diameter of the exhaust valve retainer.

©Cummins Inc.



cp8rtwa

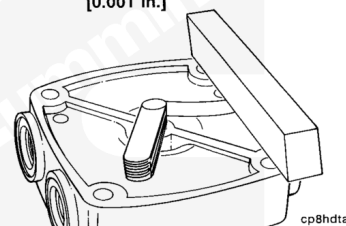
Slide the exhaust valve retainer over the valve plate. Make sure that the end of the retainer with the groove faces upward.



cp8rtha

Single-cylinder heads **must** be flat within 0.03 mm [0.001 in] between any two adjacent capscrew holes.

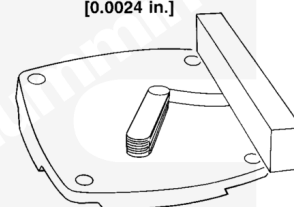
0.03 mm  
[0.001 in.]



cp8hda

Single-cylinder top cover **must** be flat within 0.06 mm [0.0024 in] between any two adjacent capscrew holes and 0.10 mm [0.004 in] total.

0.06 mm  
[0.0024 in.]



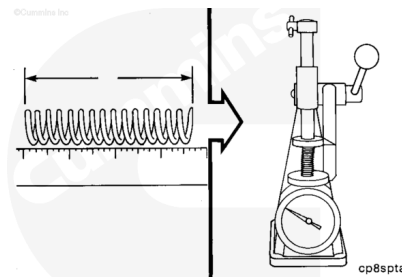
cp8cvna

Use the valve spring tester, Part Number 3375182, to check the unloader valve spring (blue stripe).

Replace any spring that does **not** meet specifications.

Compress springs to 24.9 mm [0.98 in].

| kg   |     | lb |
|------|-----|----|
| 10.4 | MIN | 23 |
| 12.5 | MAX | 27 |



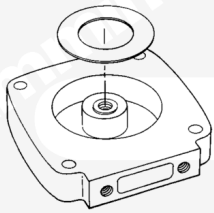
cp8spta

**Note :** Holset® recommends new springs be installed.

## Assemble

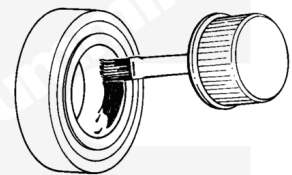
QE, Non-European

Install the exhaust valve over the post in the valve plate.



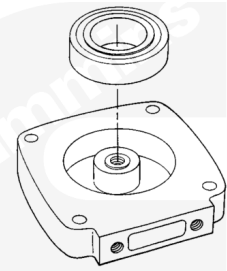
cp8vahk

Apply a thin coating of anti-seize compound to the inside circumference of the exhaust valve retainer.



cp8rtwa

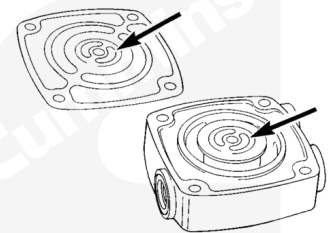
Slide the exhaust valve retainer over the valve plate. Make sure that the end of the retainer with the groove faces upward.



cp8rtha

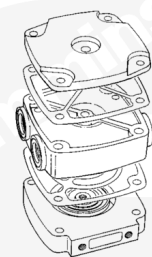
Determine the final orientation of the valve plate (air intake location) and the head (coolant ports in relation to air inlet or manifold location). Align the kidney-shaped slots in the head with the kidney-shaped slots in the gasket.

If orientation marks were made before disassembly, use them.



cp8piaa

Assemble the cover, cover gasket, head, head gasket, and valve plate.



cp8cvmg

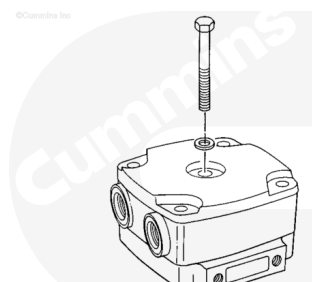
**Note :** Make sure corner capscrew holes are aligned.

Lubricate the threads under the head.



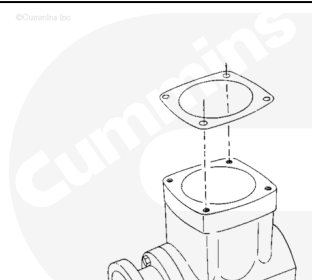
Install the shorter capscrew with washer through the center hole.

**Torque Value:** 14 n•m [124 in-lb]



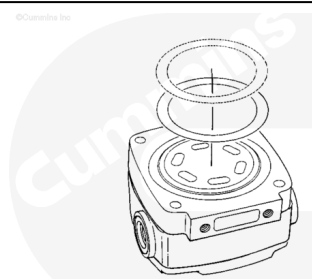
cp8hdec

Install the valve plate gasket.



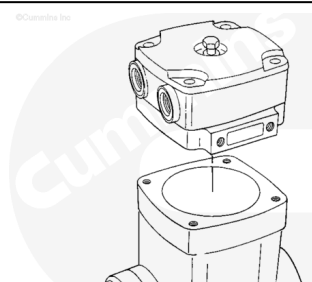
cp8pihe

Carefully place the intake valve in the valve plate. Install the intake valve retainer.



cp8vams

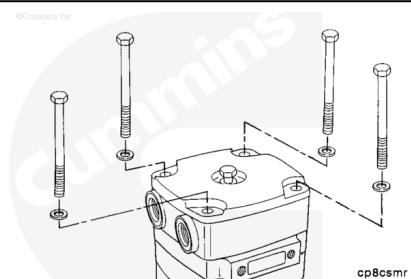
Install the valve plate assembly.



cp8hdmg

Lubricate the threads under the head and washer of the capscrews, if initially installed.

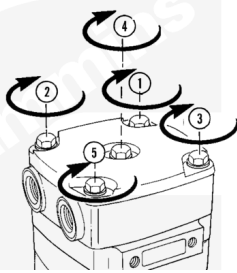
Install the four head capscrews and washers.



cp8csmr

Tighten all five capscrews in the proper sequence.

**Torque Value:** 28 n•m [21 ft-lb]

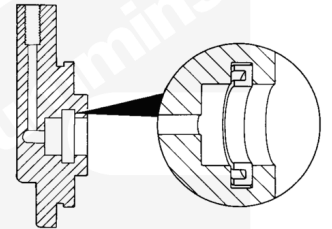


cp8cswb

Install the new rectangular v-seal into the unloader body.

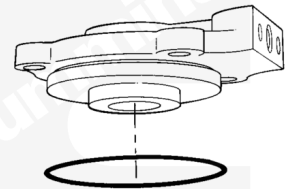
**Note :** The seal must be installed with the grooved side up.

Liberal lubricate the unloader valve bore above and below the rectangular ring seal with high-temperature grease (Accrolube lubrication Teflon™ grease or equivalent).



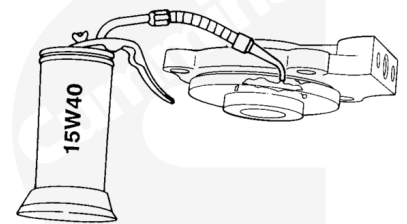
cp8sehg

Install a new o-ring seal on the unloader valve body.



cp8sehh

Use clean lubricating engine oil or Accrolube lubrication Teflon™ grease, or equivalent, to lubricate the seal.

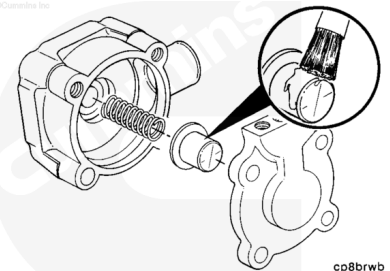


cp8sewa

Liberal lubricate the unloader valve body bore and unloader cap with high-temperature grease (Accrolube lubrication Teflon™ grease or the equivalent).

Install the unloader cap.

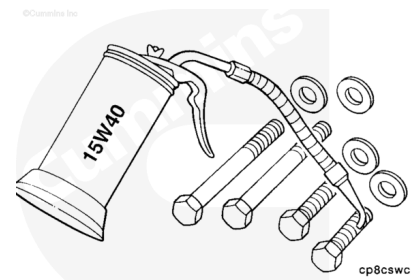
Install the unloader spring.



cp8brwb

Lubricate the unloader screw threads and underhead with clean lubricating engine oil before installation.

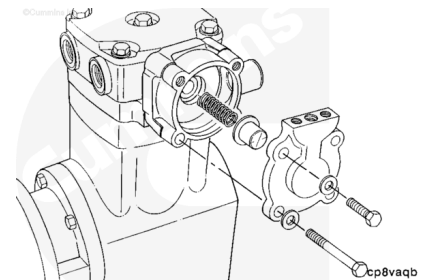
**Note :** The two unloader body screws must not be used to attach any brackets.



Assemble the unloader components, and attach the unloader assembly to the valve plate with the four capscrews and washers.

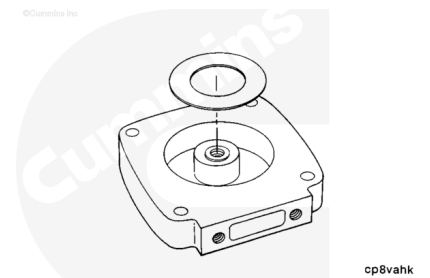
**Note :** The longer capscrews are used to mount the manifold to the air compressor.

**Torque Value:** 27 n•m [ 20 ft-lb ]

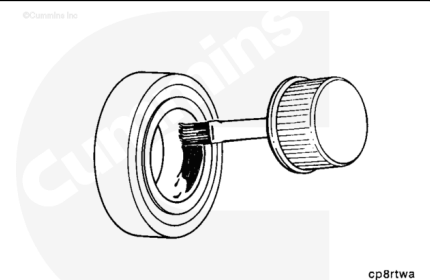


QE, European

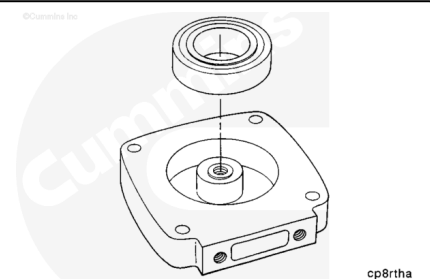
Install the exhaust valve over the post in the valve plate.



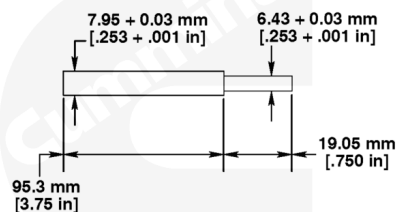
Apply a thin coating of anti-seize compound to the inside circumference of the exhaust valve retainer.



Slide the exhaust valve retainer over the valve plate. Make sure that the end of the retainer with the groove faces upward.



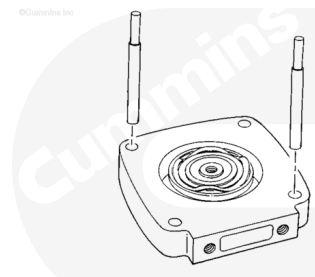
Fabricate or reuse the four guide pin tools.



cp8pixa

Insert the larger end of two guide pins in opposite corner holes of the valve plate.

©Cummins Inc.

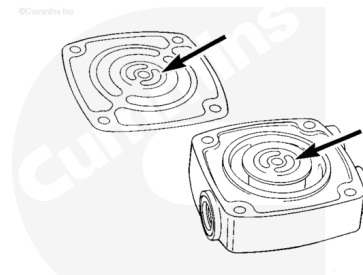


cp8pihc

Determine the final orientation of the valve plate (air intake location) and the head (coolant ports in relation to air inlet or manifold location). Align the kidney-shaped slots in the head with the kidney-shaped slots in the gasket.

If orientation marks were made before disassembly, use them.

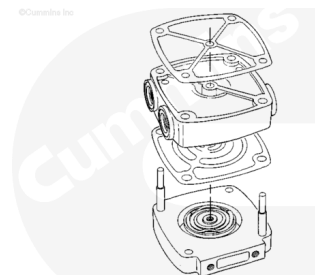
©Cummins Inc.



cp8piaa

Install the head gasket onto the guide pins (either side up, but with correct slot orientation). Install the head onto the guide pins with the kidney-shaped slots aligned and toward the valve plate. Install the cover gasket over the guide pins.

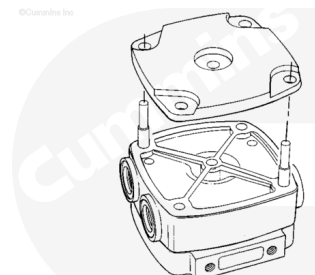
©Cummins Inc.



cp8gkhf

Assemble the cover.

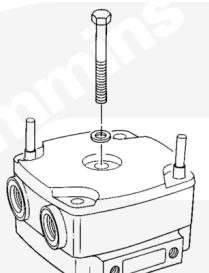
©Cummins Inc.



cp8cvhb

Install the shorter capscrew with washer, if initially installed, through the center hole.

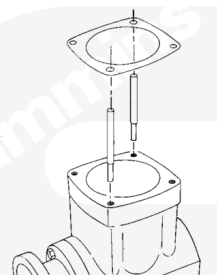
**Torque Value:** 14 n•m [124 in-lb]



cp8cshh

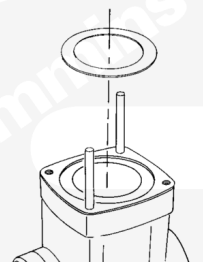
Place the remaining two guide pins in the crankcase head capscrew holes (that will **not** interfere with the guide pins already in the head assembly).

Install the valve plate gasket.



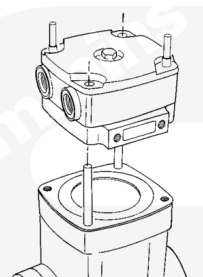
cp8pihd

Carefully place the intake valve on the crankcase, located by the valve plate gasket. Do **not** allow the valve to overlap the gasket.



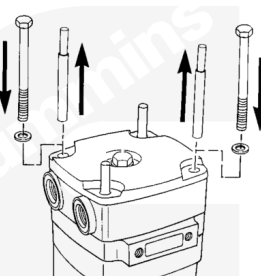
cp8vahf

Install the head assembly over the guide pins. Be careful **not** to disturb the location of the intake valve. The compressor will **not** work if the valve overlaps the gasket and is pinched.



cp8hdhb

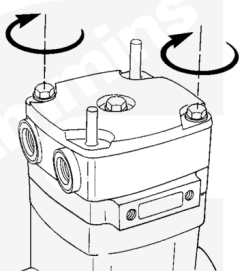
Carefully remove two of the guide pins and replace with two head capscrews and washers, if initially installed.



cp8pimb

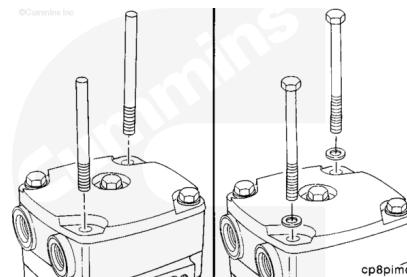
Tighten the two head capscrews.

**Torque Value:** 14 N·m [ 124 in-lb ]



cp8cswa

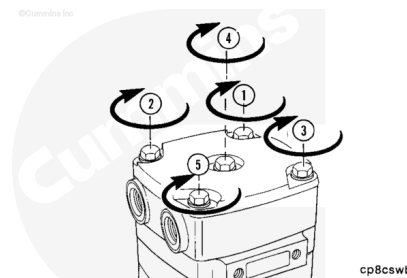
Remove the remaining two pins and replace with two head capscrews and washers, if initially installed.



cp8pimc

Tighten all five capscrews in the proper sequence.

**Torque Value:** 28 n·m [ 21 ft-lb ]

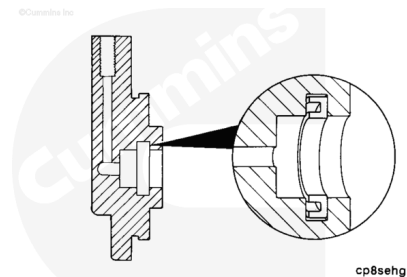


cp8cswb

Install the new rectangular v-seal into the unloader body.

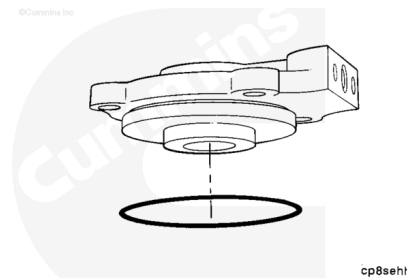
**Note :** The seal must be installed with the grooved side up.

Liberal lubricate the unloader valve bore above and below the rectangular ring seal with high-temperature grease (Accrolube lubrication Teflon™ grease or equivalent).



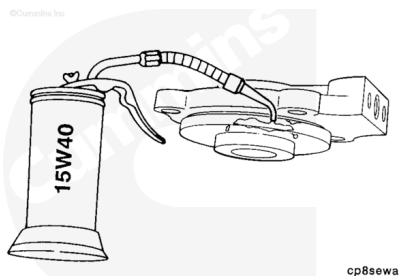
cp8sehg

Install a new o-ring seal on the unloader valve body.



cp8sehh

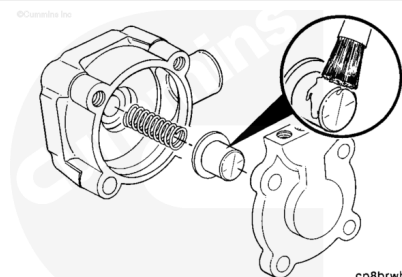
Use clean lubricating engine oil or Accrolube lubrication Teflon™ grease, or equivalent, to lubricate the seal.



Liberalily lubricate the unloader valve body bore and unloader cap with high temperature grease (Accrolube lubrication Teflon™ grease or the equivalent).

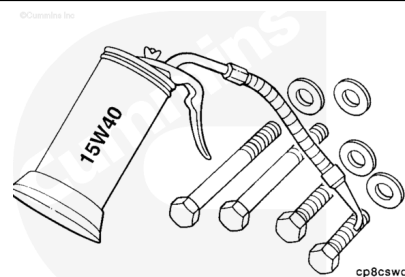
Install the unloader cap.

Install the unloader spring.



Lubricate the unloader screw threads and under the head with clean lubricating engine oil before installation.

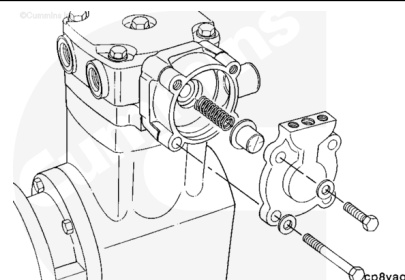
**Note :** The two unloader body screws must not be used to attach any brackets.



Assemble the unloader components, and attach the unloader assembly to the valve plate with the four capscrews and washers.

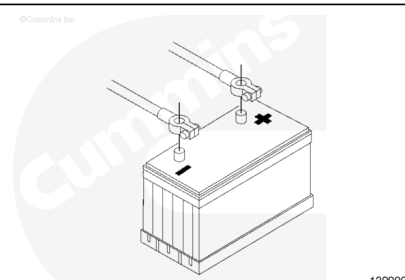
**Note :** The longer capscrews are used to mount the manifold to the air compressor.

**Torque Value:** 27 n•m [ 20 ft-lb ]



## Finishing Steps

**Note :** If the cylinder head was removed while the air compressor was on the engine, fill the engine coolant. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html).



**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

- Connect the batteries
- Operate the engine and check for leaks.